# System Driven Hardware Design

- Uint8 to Matlab -

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# Agenda

- Exercise
- Provided Matlab Script
- Resources

# Exercise 2: Uint8 Transfer to Matlab



**Implement**
Everytime you send the character ‘s’ from Matlab to the FreeSoc2, it answers with 256 uint8 values, containing a triangle from [0 to 128 and 127 to 1].

**Use**
UART Block
The provided Matlab Script

**Note**
Sequencing:

1. power on FreeSoc2 (connect to USB-Port)
2. Start Matlab
3. Run the Matlab Script

# 1/4 Provided Matlab Script: Preamble

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%
% Uint8 Datatransfer from FreeSoc2 to Matlab
% Version 1.0, Bannwarth, 21.05.2020
%
% Behaviour:
% 1. sends 's' over the serial Port
% 2. Waits until there are 256 bytes in the
buffer
% 3. If there 256 bytes in the buffer, it take
them, calculates the fft
%    out of it and displays the received data
and the absolute of the fft.
%
% Using:
% 1. Connect FreeSoc2 to USB (i.e. Power Up)
% 2. Check the correct serial Port Settings
% 3. Run this Script
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%
```

# 2/4 Provided Matlab Script: Setup

```
close all;
clear all;
clc;


% Setup and open the serial communication in Matlab
priorPorts=instrfind;
delete(priorPorts);
PSoC=serial('COM5', 'BaudRate', 9600, 'InputBufferSize',
512); % implecit: Parity off / stop bit: 1
fopen(PSoC);

% flags
flg_data_avai = 0;
rx_meas_num = 0;
```

# 3/4 Provided Matlab Script: Displaying and Ending

```
figure
plot([1:256],rx_data);
title('Received Time Domain Data');

figure
plot([1:256],1/length(rx_data)*abs(fft(rx_data)));
title('FFT - from Matlab');

fprintf(" Scipt End \n");
```

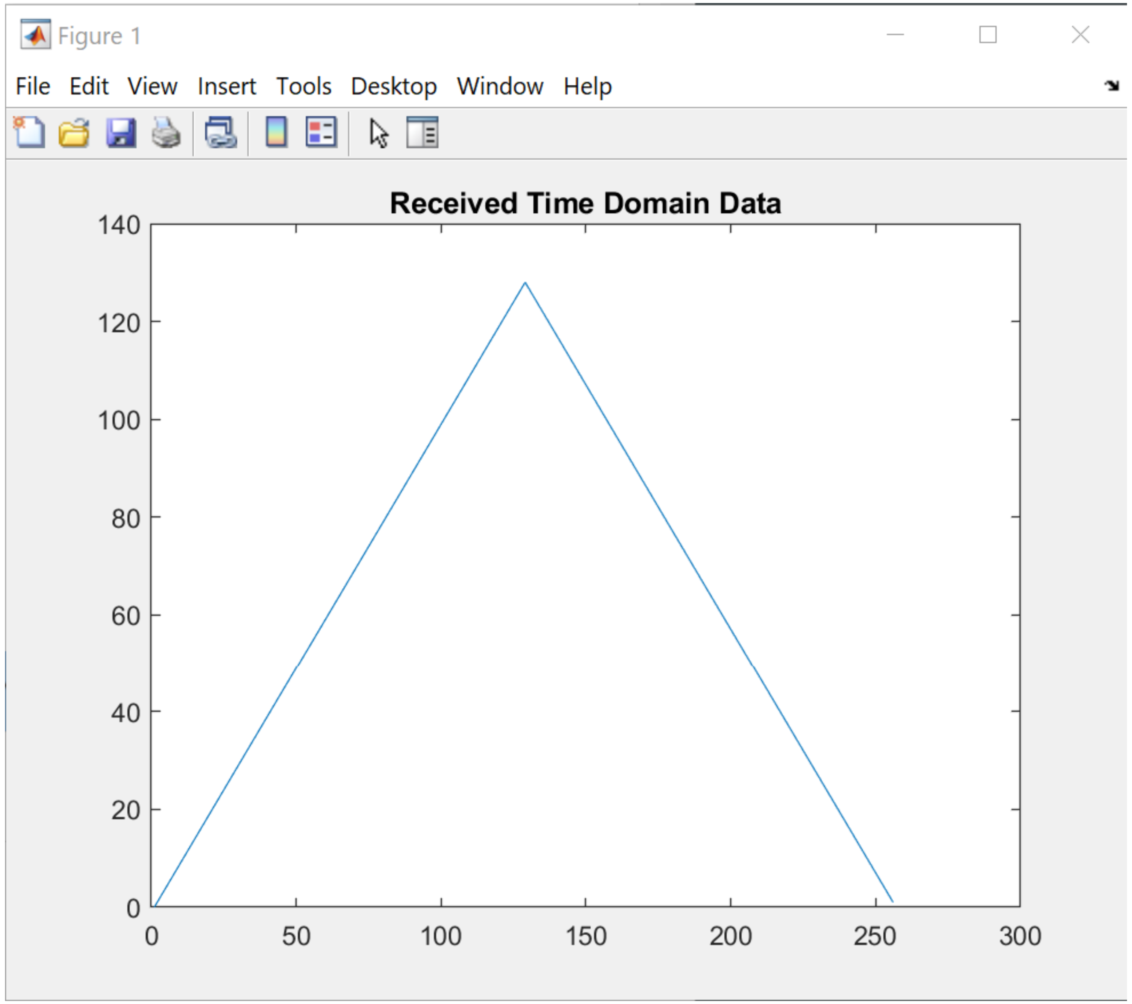
# 4/4 Provided Matlab Script: Collecting data

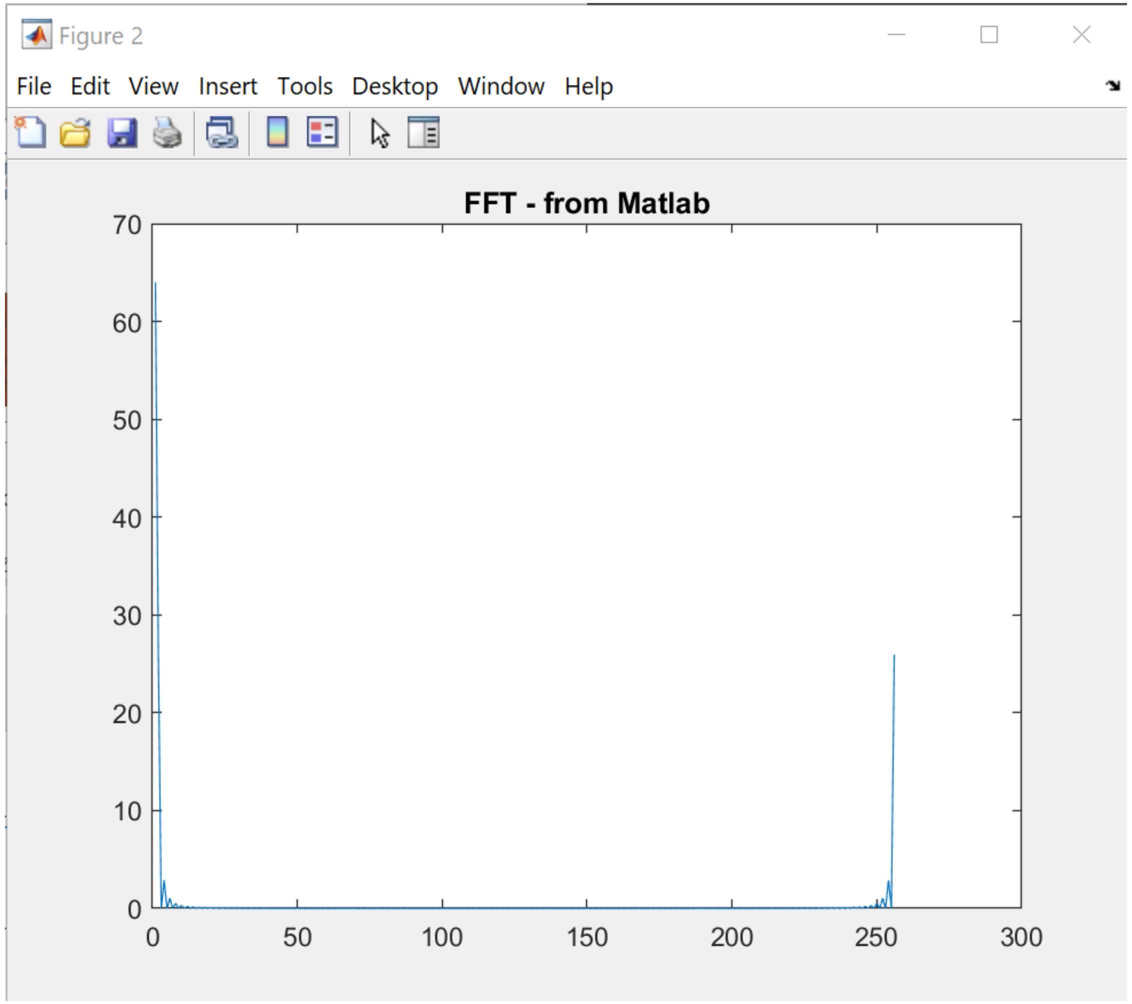


```
% Matlab ready to send: Command to PSoc to send data
fwrite(PSoC,'s','uchar') % means ready to receive, send data

while(flg_data_avai == 0)

        if PSoC.BytesAvailable == 0

            fprintf(" Waiting \n");

        end

       if PSoC.BytesAvailable == 256 % When the 256 Bytes are sent, the
buffer is read
             rx_data = fread(PSoC,256,'uint8');
             fprintf(" Transfer %i DONE \n",rx_meas_num);
             flg_data_avai = 1;
       end

end

fclose(PSoC);
```

# Result after running `Uint2Matlab.m`





**FreeSoc**

- “Universal Asynchronous Receiver Transmitter (UART)” Cypress, PSoC® Creator™ Component Datasheet
- “KitProg User Guide” see section: 5. USB UART Bridge, Cypress, Userguide

**Matlab Help**

- Serial Port Devices - Read and write to devices connected to a serial portRead and write to devices connected to a serial port

Documentation > MATLAB > Data Import and Analysis > Data Import and Export

- fread (serial)